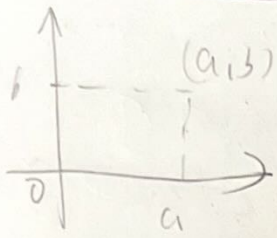
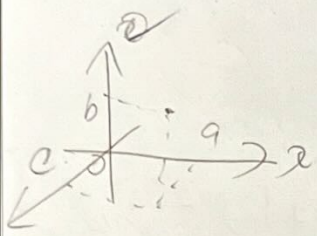
(ex2.1.15) 22054 鈴木

8.1.14 CNF を $P = (P_1 \vee P_2 \vee P_3) \wedge (\neg P_1 \vee \neg P_2) \wedge (P_1 \vee \neg P_3) \wedge (P_2 \vee P_3)$

$P_1 = T, P_2 = F, P_3 = T$ で充足可能



平面



z 空間

例題2.1

$x \in A$ のとき, $A \subset B \wedge B \subset C$ なら, あれは $x \in B \vee$ 何かの $x \in A$ は $x \in C$ の

$A \subset B \wedge B \subset C \Rightarrow A \subset C$

n 種の命題 $P_1, P_2 \cdots P_n$ の選択肢 2^n 個

8.1.12 $\neg(p \vee q) \vee (\neg p \wedge q)$

$\Leftrightarrow (\neg p \wedge \neg q) \vee (\neg p \wedge q)$

$\Leftrightarrow \neg p(\neg q \wedge q) \Leftrightarrow \neg p \cdot 1 \Leftrightarrow \neg p$

ex 1.12

CNF を $P = (P_1 \vee P_2 \vee P_3) \wedge (\neg P_1 \vee \neg P_2 \vee \neg P_3) \wedge (P_1 \vee \neg P_3)$

$C = \{P\}$ とする

8.1.13

$q = p$, $C = \phi$ a) $p \vee (p \Rightarrow q) \Leftrightarrow T$ b) $p \wedge$

$p \vee (p \Rightarrow q) \Leftrightarrow p \vee (\neg p \vee q)$

$\Leftrightarrow T \vee q \Leftrightarrow T$

I より $P_2 = T$ とし

$(P_1 \vee T \vee P_3) \wedge (\neg P_1 \vee T \vee \neg P_3) \wedge (P_1 \vee \neg P_3) \wedge T$

$= (\neg P_1 \vee \neg P_3) \wedge (P_1 \vee \neg P_3)$

II より $P_3 = T$ で $q' = (\neg P_1 \vee T) \wedge (P_1 \vee T) = T$

(後ろに出かして終了)

$\boxed{2}_2$

$\boxed{2.11}$ $A \subset B \cap B \subset C \Rightarrow A \subset C$

$x \in A$ のとき $A \subset B \cap B \subset C$ であるが

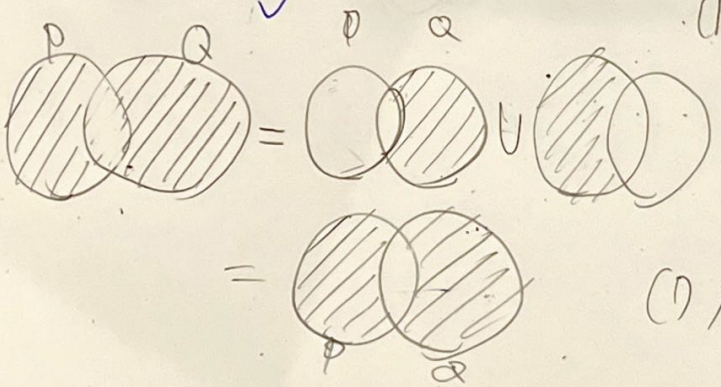
$x \in B \wedge x \in C$ よって $x \in A \wedge x \in C$

このときから成り立つ

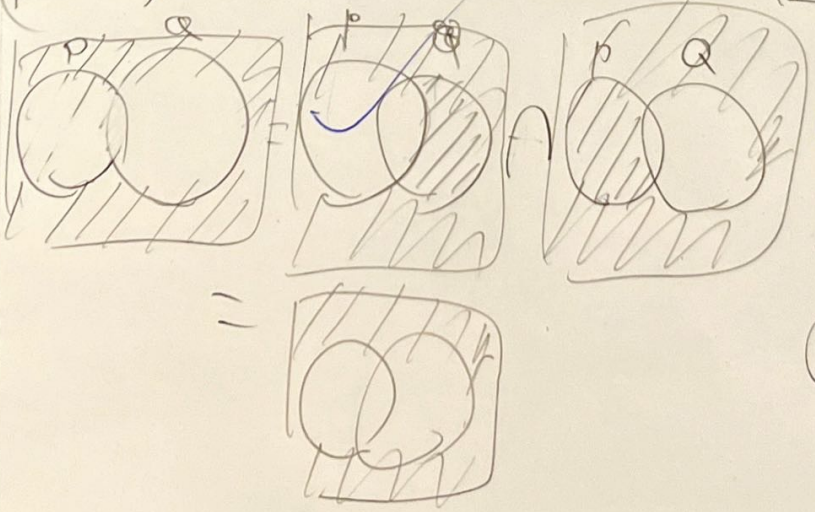
$\boxed{2.2}$ $\cap, \cup, -$ $\{2, 4, 8\}$

$(A \cap B) - (D - C)$

$\boxed{2.3}$ $(P \cap Q)^c = P^c \cup Q^c$



$(P \cup Q)^c = P^c \cap Q^c$



$\boxed{2.4}$

a) $X - Y = \{o, f, t, w, a, r, c, i, n\}$ $\{e, s, w\}$

b) $Y \cap Z = \{i, n\}$

c) $Z - (X \cup Y) = \{c\}$

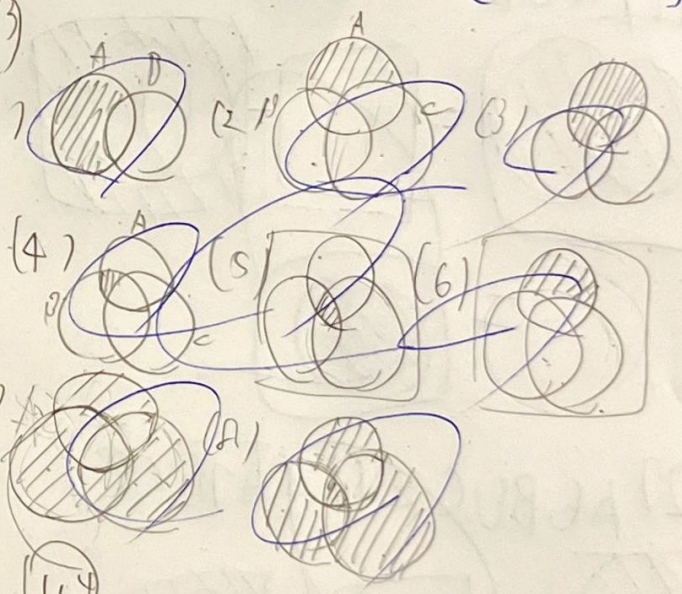
d) $X \cap Y \cap Z = \cancel{\{\emptyset\}}\ \emptyset$

e) $X^c \cap Y \cap Z^c = \cancel{\{m\}}$

f) $X^c \cap (Y^c \cup Z) = \{s, o, f, t, a, r, e\}$ $\{c, i, m, z\}$

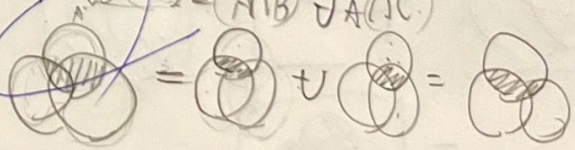
$\boxed{3}$

$\boxed{1.3}$

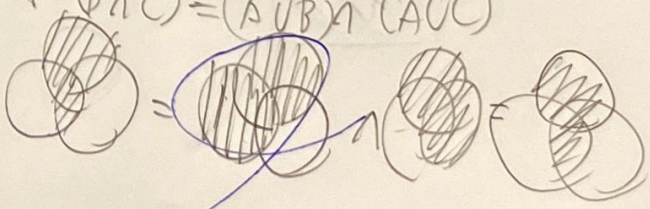


$\boxed{1.4}$

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

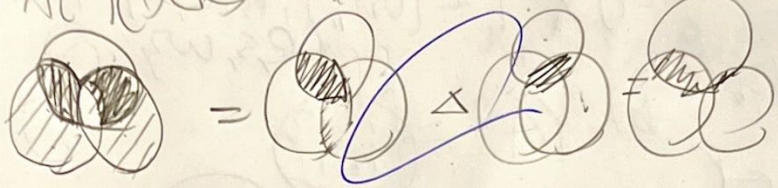


$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

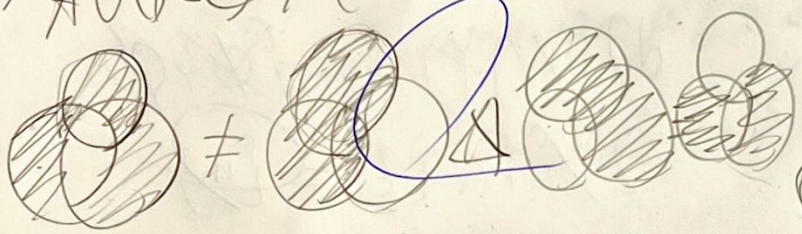


1.5

(1) $A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C)$

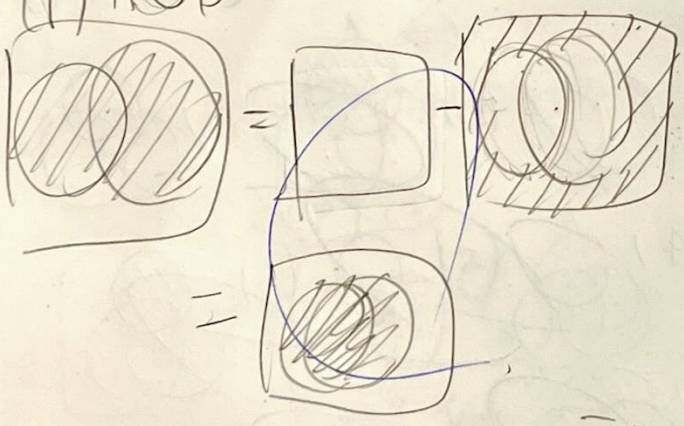


(2) $A \cup (B \triangle C) \neq (A \cup B) \cap (A \cup C)$

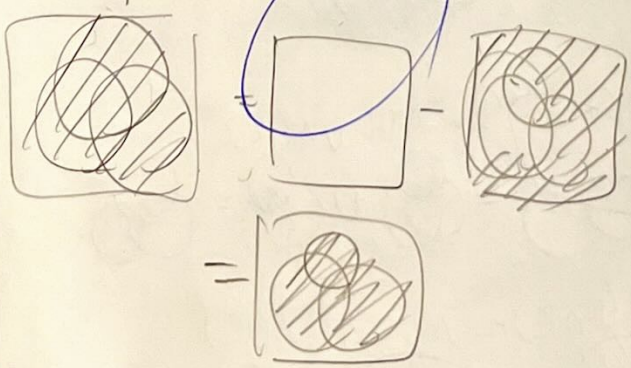


1.6

(1) $A \cup B = U - (\bar{A} \cap \bar{B})$

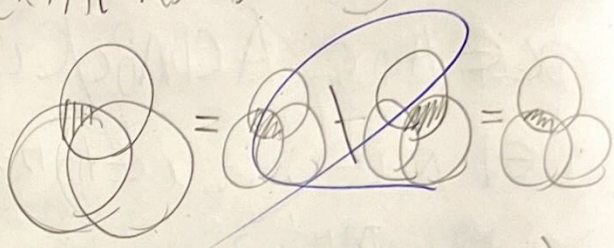


(2) $A \cup B \cup C = U - (\bar{A} \cap \bar{B} \cap \bar{C})$

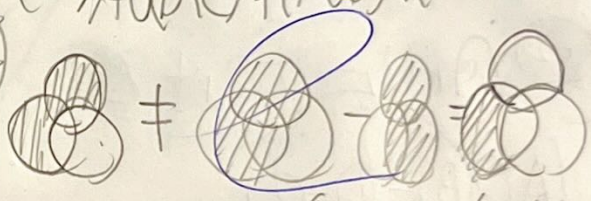
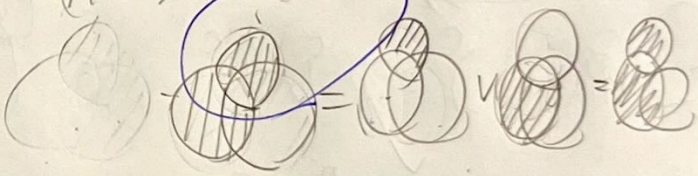


1.7

(1) $A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C)$



(2) $A \cup (B \setminus C) \neq (A \cup B) \setminus (A \cup C)$



(3) $(A \cup B) - (A \cap B) = (A \setminus B) \cup (B \setminus A)$